Flow Is Not More Than Its Parts: Decomposing the Challenge-Skill Model and Its Momentary Association with Pain in Chronic Pain

Stijn Van Severen[1,*], Ilse Viane[1], Annick De Paepe[1], and Geert Crombez[1]

[1] Department of Experimental-Clinical and Health Psychology, Ghent University, Ghent, Belgium
[*] Corresponding author: stijn.vanseveren@ugent.be

Abstract

Flow, the absorbed and intrinsically rewarding state said to arise when perceived challenge and perceived skill are both high and matched, has been proposed as a route to analgesia. The proposal rests on two structural assumptions that experience-sampling operationalizations build in rather than test: that the challenge-skill condition acts through the *match* between its two components, and that the constituent appraisals form one construct that a composite summarizes fairly. We tested both assumptions, and the momentary association with pain, in 68 patients with chronic pain who rated challenge, skill, absorption, and enjoyment together with pain, its interference, attention to pain, and its threat value eight times per day for two weeks (5,976 moments with all four appraisals, median = 92 per person). Neither assumption held. Balance, the negative absolute challenge-skill discrepancy, added nothing to an additive model of challenge and skill ($\chi^2(1) = 0.56$, $p = .45$), and neither did a one-sided kink at equality, a multiplicative interaction, or a full second-order response surface (all $p \geq .27$); between persons, better-matched activities went with *less* flow experience ($b = -.111$, $p = .028$). Entering the four constituents jointly rather than as a composite improved fit for every pain outcome (all $p < .001$), because absorption and skill tracked lower pain ($b = -.055$ and $b = -.045$) while challenge reversed sign and tracked *higher* pain ($b = +.032$, $p = .004$). The flow experience was concurrently associated with lower pain interference ($b = -.139$), pain ($b = -.094$), threat ($b = -.079$), and attention to pain ($b = -.074$, all $p < .001$), but no lag-1 effect reached significance in either direction while autoregressive effects remained large, and the least painful region of the challenge-skill plane was the low-challenge, high-skill relaxation channel rather than flow. Flow in daily life with chronic pain is best described by how absorbing and how manageable an activity is, not by how well demand and capacity are matched; the composite adds a label where the components carry the information.

Keywords: flow; challenge-skill balance; chronic pain; constructed measures; ecological momentary assessment; multilevel modelling

1 Introduction

1.1 Flow, and the proposal that it relieves pain

Flow is the state of complete absorption in an activity that is experienced as intrinsically rewarding and that arises, on the classic account, when the demands of the activity and the person's capacity to meet them are both high and in balance (Csikszentmihalyi, 1990; Nakamura and Csikszentmihalyi, 2014). The construct was built for experience sampling from the start: challenge and skill were rated at signalled moments, and the resulting challenge-skill plane was divided into four channels, flow, anxiety, relaxation, and apathy, whose quality of experience was then compared (Massimini and Carli, 1988; Carli et al., 1988; Csikszentmihalyi and Larson, 1987).

Interest in flow has recently reached pain research. Gromakovskis and Gūtmanis (2025) review evidence for flow-induced analgesia, the proposal that deep absorption in a competing activity blunts pain, and place it alongside the biopsychosocial and neurobiological accounts of pain modulation. The proposal is attractive because it converges with an established idea in pain psychology. In the cognitive-affective model of the interruptive function of pain (Eccleston and Crombez, 1999), pain and current goals compete for a limited attentional capacity, and the outcome of that competition depends on the threat value of the pain and on the characteristics of the activity it interrupts (Crombez et al., 1998, 1999; Van Damme et al., 2010). Threat-related appraisal and monitoring amplify the attentional pull of pain (Crombez et al., 2013; Eccleston and Crombez, 2007), and the same appraisals sustain avoidance of demanding activity (Crombez et al., 2012). Flow describes the opposite pole of the same competition: an activity so absorbing that competing signals fail to intrude. If that is right, flow states should be states of reduced pain.

The evidence adduced for flow-induced analgesia comes almost entirely from paradigms in which flow is induced, through virtual reality, music, or structured tasks, and pain is measured during the induction (Gromakovskis and Gūtmanis, 2025). Such designs support causal inference but say little about whether the same relation holds in the unstructured daily life of patients whose pain is persistent. That is an ecological question, and experience sampling is the method built to answer it (Myin-Germeys et al., 2018; Shiffman et al., 2008; Schwartz and Stone, 1998), with an established record in chronic pain specifically (Tennen and Affleck, 1996; Affleck et al., 1999; Peters et al., 2000).

1.2 Two structural assumptions that are usually assumed rather than tested

Before asking whether flow accompanies less pain, it is worth asking what a momentary flow score actually contains. Two structural assumptions are built into the standard operationalization.

The first is **configurality**. The channel model does not claim that high challenge is good or that high skill is good; it claims that their *correspondence* is what matters, and that correspondence at a high level is what produces flow. Operationally, this is usually captured by a balance term, the negative absolute discrepancy $-|C-S|$, entered alongside an elevation term $(C+S)/2$ that keeps a jointly low but perfectly matched moment, that is apathy, from counting as optimal (Moneta and Csikszentmihalyi, 1996). If the configural claim is right, the balance term should carry variance in the flow experience over and above what challenge and skill contribute on their own. The evidence here is mixed: meta-analytically, the challenge-skill balance relates to flow only modestly (Fong et al., 2015), and experimental work has found the balance effect to be moderated rather than universal (Engeser and Rheinberg, 2008). A further problem is methodological rather than substantive. A term built from an absolute difference imposes constraints that are rarely examined and is only interpretable if a point on the challenge scale means the same thing as a point on the skill scale; the recommended alternative is to keep the two components in their own metric and to test the congruence hypothesis as a constrained case of a polynomial response surface (Edwards and Parry, 1993; Edwards, 1994).

The second is **composite coherence**. Momentary flow is typically indexed by summing or averaging its constituent appraisals. VanderWeele (2022) shows why a constructed measure of this kind can mislead. The same composite coefficient is produced by three very different worlds: one in which all constituents relate to the outcome similarly, so that the composite is a fair summary; one in which a single constituent does all the work, so that the composite is a diluted version of that constituent under a more ambitious name; and one in which constituents work in opposite directions and partly cancel, so that the composite actively hides the structure it claims to summarize. The three cannot be distinguished from the composite itself, and the practical implication differs sharply between them, because an intervention has to know which component to move. The remedy is to report the constituents separately as well as jointly.

Both assumptions matter for the pain question specifically. Challenge, in a population with chronic pain, is not a neutral quantity. Demanding activity is also physically effortful, and effort has its own cost; a literature on activity pacing, avoidance, and endurance describes exactly this trade-off (Andrews et al., 2012; Vlaeyen and Linton, 2000). If challenge and absorption pull in opposite directions with respect to pain, a composite that averages them will be quiet about the most clinically consequential part of the construct.

1.3 The present study

This study uses a chronic-pain experience-sampling dataset in which patients answered a short momentary diary eight times per day for two weeks. Four of the diary items map onto the flow model: challenge and skill form the antecedent condition, and absorption and enjoyment form the experience itself. The same prompts recorded momentary pain, the extent to which pain interfered with what the person was doing, attention to pain, and the threat value of pain, which makes it possible to ask the ecological counterpart of the flow-induced-analgesia question. The data are a secondary analysis: the source study analysed these items separately and never as a flow index (Viane et al., 2004), so every flow variable here is an analyst-constructed measure and is reported as such. A companion paper uses the same diary to model the within-person threat-attention-pain circuit and treats the absorption item as goal engagement without flow framing; the present paper is about the flow construct itself. Because the data are observational, all associations are read as within-person predictive relations rather than causal effects (Rohrer, 2018). Three research questions guide the analysis.

RQ1. Does the challenge-skill condition relate to the momentary flow experience configurationally, as the channel model requires, or additively?

RQ2. Does the flow composite carry information about the momentary experience of pain that its four constituents, entered separately, do not?

RQ3. Is the momentary flow experience associated with a better experience of pain in daily life, concurrently and prospectively, and is the flow channel the least painful region of the challenge-skill plane?

2 Method

2.1 Participants and design

The data are a secondary analysis of a chronic-pain experience-sampling study reported by Viane et al. (2004), which examined acceptance of pain in relation to attention to pain and engagement with daily activities (see also Viane et al., 2003); the flow question asked here was not part of that report. Patients with chronic musculoskeletal pain, recruited through a university pain clinic, first completed a battery of validated questionnaires and then carried a palmtop computer that signalled them eight times per day for two weeks to complete a short momentary diary, together with a brief morning and evening diary. Sixty-eight participants met the study's compliance criterion of at least 25 completed prompts. Of their completed prompts, 5,976 carried all four activity appraisals and form the analytic frame for this paper (median = 92 per person); the 4.6% of prompts lost to a missing appraisal are characterized in Figure S1 and Table S10. Baseline questionnaires were available for 67 participants. Sample characteristics and the momentary appraisals are summarized in Table 1.

Table 1. Sample and the momentary appraisals that operationalize flow.

Panel A. Baseline characteristics of the analytic sample.

Variable	Mean (SD)	Range	*n*
Age (years)	46.4 (9.8)	21 to 65	67
Sex: women / men (%)	80.6 / 19.4		67
Pain duration (months)	82.8 (121.4)	2 to 520	62
MPI pain severity	4.3 (0.9)	2 to 6	67
MPI interference	4.2 (1.0)	2 to 6	67
Pain Disability Index	42.7 (11.8)	11 to 68	67
PCS catastrophizing (total)	22.3 (10.2)	1 to 46	66
PVAQ pain vigilance (total)	43.7 (12.8)	5 to 66	65
HADS anxiety	7.7 (3.9)	1 to 20	66
HADS depression	8.7 (3.7)	1 to 16	66
NEO-FFI neuroticism	30.8 (9.4)	12 to 54	64

Panel B. The four momentary appraisals and the flow experience composite.

Momentary item	Role	*n* obs	Mean	SD	ICC	Within-person share
Challenge	Condition	6,165	3.28	1.98	0.242	0.758
Skill	Condition	6,108	4.79	1.55	0.226	0.774
Absorption	Experience	6,143	4.00	1.90	0.213	0.787
Enjoyment	Experience	6,136	4.90	1.58	0.207	0.793
Flow experience	Experience	6,035	4.45	1.51	0.201	0.799

Note. MPI = West Haven-Yale Multidimensional Pain Inventory; PDI = Pain Disability Index; PCS = Pain Catastrophizing Scale; PVAQ = Pain Vigilance and Awareness Questionnaire; HADS = Hospital Anxiety and Depression Scale; NEO-FFI = NEO Five-Factor Inventory. Items were rated 1 (not at all) to 7 (very much). Condition items are challenge and skill, experience items absorption and enjoyment; the flow experience is their mean. ICC is the between-person variance share. The four appraisals are single momentary items; every flow variable is a secondary, analyst-constructed measure, since the source study analysed the items separately.

2.2 Momentary measures

At each prompt, participants described the activity they were engaged in and rated it on seven-point scales (1 = not at all, 7 = very much). Four ratings operationalize flow. Challenge was "this activity is a personal challenge for me" and skill was "I am good at this"; together they form the antecedent condition. Absorption was "I am absorbed in this activity" and enjoyment was "I like doing this"; together they form the experience, absorption supplying the concentration component and enjoyment the autotelic marker. The same prompt recorded momentary pain ("I am in pain now"), pain interference ("my pain hinders me"), attention to pain ("I focus on my pain"), and the threat value of pain, scored as the mean of momentary fear ("I am afraid of the pain now") and catastrophizing ("I feel that my pain is becoming too much for me"). A self-reported physical activation item was recorded as well and is used as a covariate throughout, because challenge co-varied strongly with being physically active ($r = .57$ within persons) and activity has its own relation to pain.

These four appraisals cover three of the nine dimensions Csikszentmihalyi describes, namely the challenge-skill configuration, concentration through absorption, and the autotelic quality through enjoyment. Clear goals, unambiguous feedback, sense of control, loss of self-consciousness, and time transformation were not measured. The instrument is therefore a core flow proxy rather than a flow scale (Jackson and Eklund, 2002), and it is described as such throughout.

2.3 Baseline measures

Baseline questionnaires provided three person-level profiles used in the person-level analyses. The threat-value profile combined the Pain Catastrophizing Scale total (Sullivan et al., 1995) and the Pain Vigilance and Awareness Questionnaire total (Roelofs et al., 2003). The biomedical profile combined the West Haven-Yale Multidimensional Pain Inventory pain-severity subscale (Kerns et al., 1985) and pain duration. The personality profile was NEO-FFI neuroticism (Costa and McCrae, 1992). Each profile was standardized.

2.4 Operationalizing flow

Following the standard experience-sampling operationalization (Massimini and Carli, 1988; Moneta and Csikszentmihalyi, 1996), the four items were represented as two structurally distinct parts rather than as one sum, because summing them conflates the antecedent condition with the outcome experience. The condition was expressed as balance and elevation, and the experience as the mean of absorption and enjoyment:

$$\text{balance} = -|C - S|, \qquad \text{elevation} = \tfrac{1}{2}(C + S), \qquad \text{flow experience} = \tfrac{1}{2}(A + E), \tag{1}$$

with C challenge, S skill, A absorption, and E enjoyment. The elevation term is not optional: without it, a moment low on both challenge and skill would score as perfectly balanced and therefore as optimal, which is apathy rather than flow. For the descriptive channel analysis a moment was assigned to flow when both condition items were high, to anxiety when only challenge was high, to relaxation when only skill was high, and to apathy when neither was (Carli et al., 1988).

Two measurement decisions shape everything that follows and are reported explicitly. The first is the metric. Person-mean centering was primary (Curran and Bauer, 2011), because it separates a person's momentary fluctuation from their level. Within-person standardization was run as a sensitivity analysis but cannot carry prevalence claims: it sets every participant's mean to zero and therefore assigns above-average moments to everyone, including a participant who never reaches flow in any absolute sense. All prevalence statements are therefore made on the raw absolute scale, with both a conservative criterion (both condition items at least 5) and a liberal one (both at least 4); the grand-mean z metric was run as a further sensitivity analysis (Figure S2, Table S2, Table S3). The second is provenance: these are single momentary items, not validated subscales, and the source study analysed them separately, so every flow variable is a secondary, analyst-constructed measure.

2.5 Statistical modelling

Analyses were implemented in a reproducible pipeline in which statistical models were written in R (R Core Team, 2024) and orchestrated from Python, which handled preprocessing, tables, and visualization. Multilevel models were fitted with lme4 (Bates et al., 2015). Every reported table and figure was regenerated from the processed data.

RQ1: configural or additive. The condition-to-experience model regressed the flow experience on balance and elevation, decomposed into within-person and between-person components, with random slopes for both within-person terms. Because the balance parameterization is only one of several ways to express a configural claim, and because a term built from an absolute difference is interpretable only under a commensurability assumption that two single items cannot guarantee (Edwards and Parry, 1993; Edwards, 1994), the configural hypothesis was tested in four further forms, each of which keeps challenge and skill in their own metric and each of which is a nested extension of the additive model: a balance term added to the additive model, a one-sided kink at $C = S$ that allows overload to differ from boredom, a multiplicative $C \times S$ interaction, a full second-order response surface, and the four-channel quadrant classification. These nested comparisons were estimated by maximum likelihood with a random intercept, so that only the fixed-effect structure differs between the compared models, and were evaluated by likelihood-ratio test, AIC, and BIC. The two parameterizations of the condition, balance-and-elevation versus additive challenge-and-skill, have the same number of parameters and are not nested, so they were compared by information criterion.

RQ2: does the composite carry the information. Absorption, enjoyment, challenge, and skill were entered as the only appraisal in the model ("alone") and then all four together ("jointly"), predicting each of the four momentary pain measures, with physical activation as a covariate. The comparison of the two sets is the constituent decomposition that VanderWeele (2022) recommends, and a change of sign between them is the signature of statistical suppression (MacKinnon et al., 2000). Because the flow experience is the mean of absorption and enjoyment, the composite model is exactly the constituent model under the constraint that those two carry equal weight and that challenge and skill carry none, so the two are nested and can be compared by likelihood-ratio test. As an independent, multivariate check, the companion paper's benchmark multilevel vector autoregressive network (Epskamp et al., 2018) was re-estimated with the flow composite substituted for the single absorption node, and the two networks were compared edge by edge.

RQ3: flow and the momentary experience of pain. Concurrent multilevel models regressed each pain measure on the flow experience, with the condition terms in the model, a random slope for the flow experience, and both without and with physical activation as a covariate. Temporal models regressed each pain measure at t on the flow experience at $t-1$ while controlling the outcome's own autoregression, and the reverse direction as well, so that the temporal picture is reported in both directions rather than in one. Lags were built between adjacent prompts within the same study day, so no lag crosses the overnight gap, and were estimated on person-standardized scores. The four channels were compared with apathy as the reference, the conventional contrast, and again with flow as the reference, which turns the ordering claim into a direct test. Person-specific associations were estimated by unregularized ordinary least squares for each participant with at least 50 usable moments.

Robustness. Every focal model was re-estimated on all four standardization metrics, with day and prompt position as level-1 covariates, and at compliance floors of 20, 30, 50, and 70 completed moments per person. Moments lost to listwise deletion were compared with retained moments on every momentary measure. No correction for multiplicity was applied; the nested tests are pre-specified contrasts of a single construct rather than a screen.

2.6 Hypotheses

Six hypotheses follow from the flow literature and from the flow-induced-analgesia proposal. For RQ1, H1 predicted configurality: balance should relate positively to the flow experience over and above elevation, and more generally some configural term should improve on the additive model. For RQ2, H2 predicted composite coherence: the four constituents should relate to pain in the same direction and to a comparable degree, so that entering them separately adds nothing. For RQ3, H3 predicted a concurrent association, with the flow experience accompanying lower pain and, more strongly, less pain interference, since an attentional account targets the interruptive rather than the sensory aspect of pain; H4 predicted the temporal counterpart, that a higher than usual flow experience would be followed by lower pain at the next prompt; and H5 predicted that the flow channel would be the least painful of the four channels. H6 predicted that these associations would vary substantially across patients.

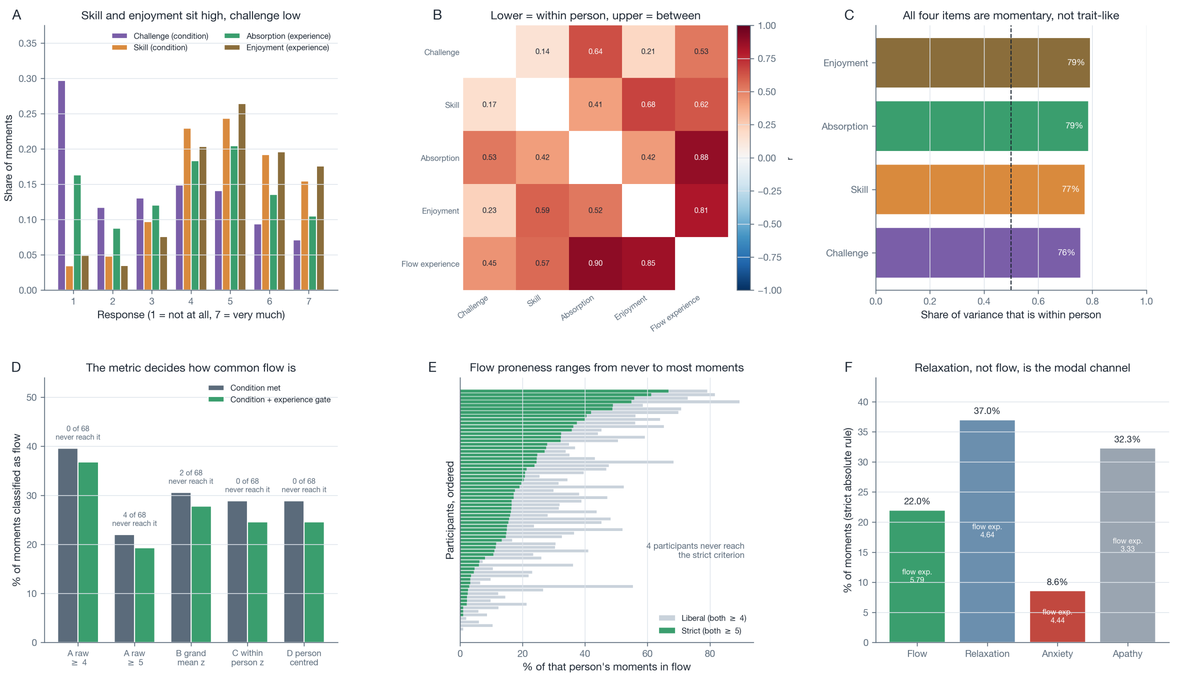
3 Results

3.1 The instrument, and how often it finds flow

The four appraisals behaved as momentary states rather than as traits: between 76 and 79% of each item's variance was within person (ICCs .21 to .24; Figure 1, Panel C, Table 1). Their levels were not symmetric. Skill (mean = 4.79) and enjoyment (mean = 4.90) sat well above the scale midpoint while challenge sat well below it (mean = 3.28), so most everyday activity in this sample was manageable and liked but not demanding (Figure 1, Panel A). Within persons, challenge correlated only $r = .17$ with skill, whereas absorption correlated $r = .53$ with challenge and $r = .42$ with skill (Figure 1, Panel B, Table S1).

How often the diary calls a moment a flow moment depended almost entirely on the classification rule (Figure 1, Panel D, Table S2). Under the conservative absolute criterion, 19.3% of moments qualified and four of 68 participants never reached the criterion at all; under the liberal absolute criterion 36.8% qualified and no participant scored zero. The relative rules gave intermediate figures (24.6 to 27.8%) but, as expected, placed no participant at zero, which is why they cannot carry a prevalence claim. Person-level proneness ranged from never to two thirds of a participant's moments (Figure 1, Panel E). One descriptive result is worth stating before any modelling: under the conservative rule the modal channel was not flow (22.0% of moments met the condition) but relaxation, that is high skill with low challenge (37.0%), with apathy next (32.3%) and anxiety rare (8.6%; Figure 1, Panel F).

Figure 1. The four momentary appraisals and how often the diary calls a moment a flow moment.



Note. Panel A shows the response distribution of the four appraisals, labelled by their role in the flow model. Panel B shows their correlation structure, within persons below the diagonal and between persons above it. Panel C shows the share of each item's variance that lies within persons. Panel D shows the share of moments classified as flow under each rule, before and after the experience gate, annotated with the number of participants who never reach the criterion. Panel E shows the per-person share of flow moments under the liberal and the strict absolute criterion. Panel F shows the share of moments in each channel under the strict rule, annotated with the mean flow experience of that channel.

3.2 RQ1: the condition acts additively, not configurationally

In the standard parameterization, the challenge-skill condition predicted the flow experience strongly, and did so entirely through elevation (within persons $b = .744$, $SE = .024$, $p < .001$; between persons $b = .796$, $SE = .085$, $p < .001$) rather than through balance (within persons $b = .006$, $SE = .020$, 95% CI $[-.033, .045]$, $p = .78$; Figure 2, Panel A, Table 2). With just under 6,000 moments this is an informative null rather than an absence of power: the interval rules out any balance effect larger than .05, against an elevation effect of .74 on the same scale (Lakens et al., 2018). Between persons the balance effect was not merely absent but reversed, so that participants whose activities were on average better matched reported on average *less* flow experience ($b = -.111$, $SE = .049$, $p = .028$). The reversal was not an artefact of centering: on the raw response scale it was larger and more precisely estimated ($b = -.090$, $p < .001$), and across all four standardization metrics the elevation effect stayed between .74 and .89 while the balance effect was never positive (Figure 2, Panel B, Table S3). Person by person, the balance slope was centred on zero (median = +.009, range −.27 to +.31) whereas the elevation slope was positive in all 68 participants (median = +.758, range +.23 to +.98; Figure 2, Panel F).

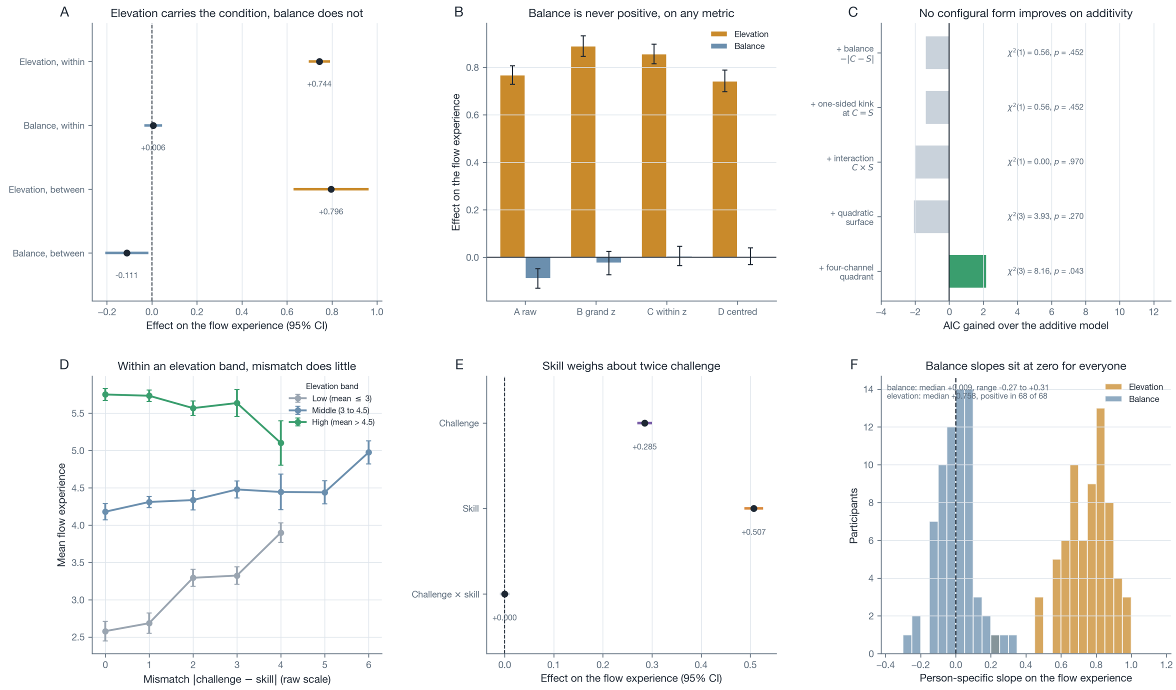
Because the balance term expresses only one of several possible configural claims, and because it is interpretable only if a point on the challenge scale means the same as a point on the skill scale (Edwards and Parry, 1993), the configural hypothesis was tested in four further forms, each keeping challenge and skill in their own metric and each nested in a comparison with the additive model (Figure 2, Panel C, Table 2). None improved on additivity. Adding balance to a model that already contained challenge and skill gained nothing ($\chi^2(1) = 0.56$, $p = .45$, $\Delta\text{AIC} = -1.4$). Allowing a one-sided kink at $C = S$, so that overload could differ from boredom, gained nothing ($\chi^2(1) = 0.56$, $p = .45$). A multiplicative $C \times S$ interaction gained nothing ($\chi^2(1) = 0.00$, $p = .97$), as it also did not in the random-slope model reported in Table S4 ($p = .15$). A full second-order response surface gained nothing ($\chi^2(3) = 3.93$, $p = .27$). The four-channel quadrant classification was the only configural form to reach conventional significance ($\chi^2(3) = 8.16$, $p = .043$, $\Delta\text{AIC} = +2.2$), but BIC preferred the additive model over it by 17.9, so the quadrant scheme buys a marginal improvement at a cost that the more conservative criterion rejects. The additive model also decisively outperformed the balance-and-elevation form at the same number of parameters ($\Delta\text{AIC} = 265.7$ with a random intercept, and 441.1 in the random-slope models of Table 2), which is what should happen when a parameterization discards information: the balance form constrains challenge and skill to equal weight, whereas the free additive estimates put skill at roughly twice challenge ($b = .507$ versus $b = .285$; Figure 2, Panel E).

Two comparisons in which balance does appear to matter are reported for completeness and are diagnostics of the parameterization rather than evidence for configurality (Figure S3, Table S5). Balance interacts with elevation inside the balance-and-elevation form ($\chi^2(1) = 17.6$, $p < .001$), with a balance effect that is negative at low elevation ($b = -.031$, $p = .06$) and positive at high elevation ($b = +.070$, $p < .001$); this is what happens when a parameterization forces two components with unequal weights into a single symmetric difference, and it disappears once challenge and skill are free. Balance also gains significance when added to the full quadratic surface ($\chi^2(1) = 18.87$, $p < .001$), but balance correlates .93 with the squared discrepancy that the surface already contains, so neither coefficient is interpretable alone, and the joint test of the surface against additivity is null. The most direct evidence is descriptive and needs no parameterization at all: within a band of elevation, mean flow experience is flat or slightly declining across the whole range of mismatch (Figure 2, Panel D), and the residuals of the additive model show no ridge along the $C = S$ diagonal (Figure S3, Panel A). H1 was therefore not supported.

3.3 RQ2: the composite hides a sign reversal among its constituents

Entered as the only appraisal in the model, three of the four constituents tracked lower momentary pain: skill ($b = -.088$), enjoyment ($b = -.077$), and absorption ($b = -.069$, all $p < .001$), while challenge was flat ($b = -.004$, $p = .82$). The composite behaved as their average ($b = -.096$, $p < .001$). Entered jointly, the picture changed in three ways (Figure 3, Panels A and B, Table 3). Absorption ($b = -.055$, $p < .001$) and skill ($b = -.045$, $p < .001$) kept their associations. Enjoyment fell to non-significance

Figure 2. Is the challenge-skill condition configural or additive?



Note. Panel A shows the balance and elevation effects on the flow experience, within and between persons. Panel B shows the same two terms estimated on each of the four standardization metrics. Panel C shows the AIC gained over the additive challenge-plus-skill model by each configural extension, annotated with its likelihood-ratio test; green marks $p < .05$. Panel D shows the observed mean flow experience against the size of the challenge-skill mismatch within three bands of elevation. Panel E shows the additive main effects and the multiplicative interaction. Panel F shows the person-specific balance and elevation slopes. Error bars and intervals are 95% confidence intervals.

($b = -.023$, $p = .089$), so its apparent relation to pain was variance it shared with the other two rather than a contribution of its own. And challenge reversed sign and became reliably positive ($b = +.032$, $p = .004$): once skill and absorption are held constant, a more demanding moment is a more painful one. The reversal is the classic signature of suppression (MacKinnon et al., 2000), and its source is visible in the correlation structure, since challenge co-varies $r = .53$ with absorption and $r = .17$ with skill, both of which track lower pain (Figure 3, Panel E). It was not confined to pain intensity: challenge was reliably positive for pain interference ($b = +.053$), attention to pain ($b = +.031$), and the threat value of pain ($b = +.035$, all $p < .01$), so on every outcome one constituent of the flow construct works against the other three (Figure 3, Panel B).

Because the flow experience is the mean of absorption and enjoyment, the composite model is the constituent model under a linear constraint, and the two can be compared directly. For pain intensity, splitting the composite into its two experience components changed nothing ($\chi^2(1) = 0.02$, $p = .89$), so the composite is a fair summary of *those two*. Adding the two condition items, however, improved fit substantially for every outcome (pain $\chi^2(3) = 21.0$, interference $\chi^2(3) = 36.1$, attention $\chi^2(3) = 26.5$, threat $\chi^2(3) = 34.0$, all $p < .001$; Figure 3, Panel D). The flow architecture as a whole therefore loses information relative to its parts, and it loses it in the specific way VanderWeele (2022) warns about, by averaging over constituents that pull in opposite directions.

An independent, multivariate check pointed the same way. Substituting the flow composite for the single absorption node in the companion paper's four-node temporal network reproduced that network

Table 2. Is the challenge-skill condition configural or additive?

Panel A. Balance and elevation predicting the flow experience (multilevel model with random slopes).

Term	b	SE	95% CI	p	Random SD
Balance, within person	0.006	0.020	[-0.033, 0.045]	.782	0.131
Elevation, within person	0.744	0.024	[0.697, 0.791]	<.001	0.164
Balance, between persons	-0.111	0.049	[-0.207, -0.015]	.028	
Elevation, between persons	0.796	0.085	[0.629, 0.963]	<.001	

Panel B. Nested likelihood-ratio tests of every configural functional form, each added to the additive challenge-plus-skill model.

Term added to challenge + skill	Δ df	χ^2	p	Δ AIC	Δ BIC
Balance, $-\lvert C-S\rvert$	1	0.56	.452	-1.4	-8.1
One-sided kink at $C = S$	1	0.56	.452	-1.4	-8.1
Interaction, $C \times S$	1	0.00	.970	-2.0	-8.7
Quadratic response surface	3	3.93	.270	-2.1	-22.2
Four-channel quadrant	3	8.16	.043	2.2	-17.9

Panel C. The two parameterizations of the condition at equal degrees of freedom.

Parameterization	df	AIC	BIC	Δ AIC
balance + elevation	5	17545.4	17578.8	265.7
challenge + skill (additive)	5	17279.7	17313.2	0.0

Note. Balance is $-\lvert C-S\rvert$ and elevation $(C+S)/2$, both person-mean centred; C is challenge and S skill. Panel A is estimated with REML and random slopes for both within-person terms. Panels B and C are estimated with maximum likelihood and a random intercept, so that only the fixed-effect structure differs across the compared models. Positive Δ AIC and Δ BIC favour the more complex model. The four appraisals are single momentary items; every flow variable is a secondary, analyst-constructed measure, since the source study analysed the items separately.

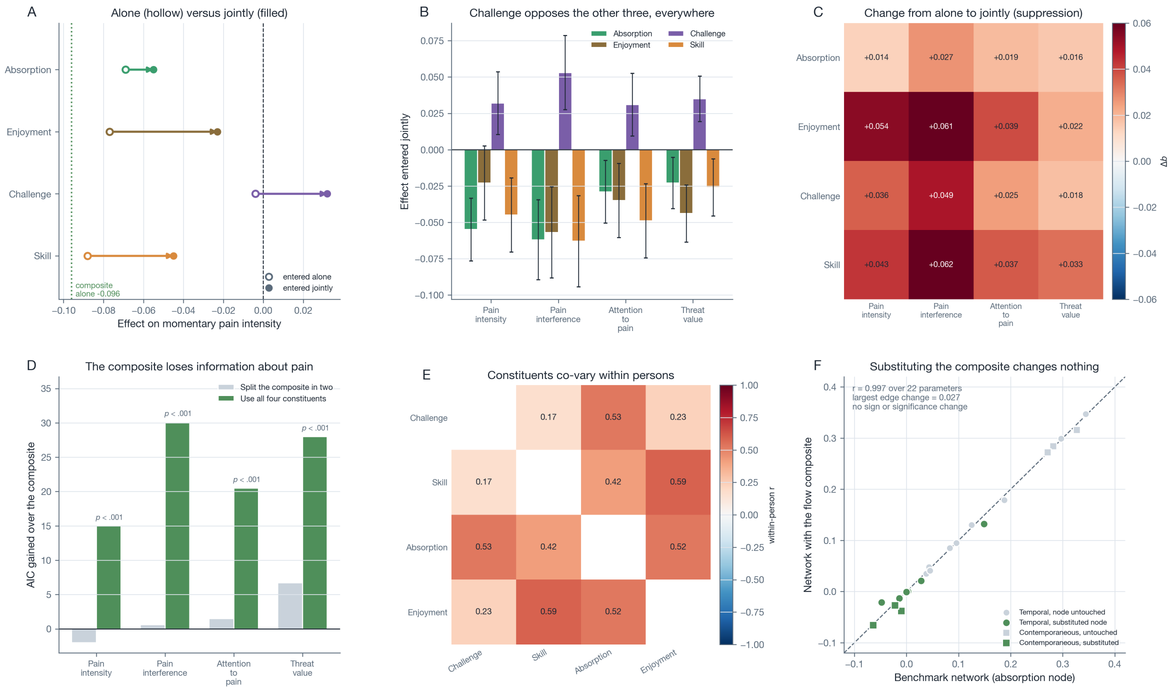
almost exactly: across all 22 estimated parameters the two networks correlated $r = .997$, the largest edge change was .027, and no edge changed sign or significance (Figure 3, Panel F, Table S7). In a dynamic system of four nodes, then, the composite behaves indistinguishably from its dominant constituent. H2 was not supported.

3.4 RQ3: a concurrent association, no temporal one, and a channel ordering that inverts the prediction

At the same prompt, a higher than usual flow experience accompanied a better experience of pain on all four measures: less pain interference ($b = -.139$, $SE = .025$), lower pain intensity ($b = -.094$, $SE = .020$), a lower threat value ($b = -.079$, $SE = .014$), and less attention to pain ($b = -.074$, $SE = .019$; all $p < .001$; Figure 4, Panel A, Table 4). All four survived adjustment for physical activation, which matters because challenge co-varies with being active and activity independently raises pain in this sample (Figure S6, Panels A and B). The ordering of the effects is itself informative and is what an attentional account predicts: interference, the interruptive aspect of pain, moved most, and moved half again as much as intensity, the sensory aspect. H3 was supported.

The temporal test gave the opposite answer. No lag-1 effect approached significance in either direction: neither the flow experience at $t-1$ predicting any pain measure at t (p between .44 and .92), nor any pain measure at $t-1$ predicting the flow experience at t (p between .28 and .94), with every point estimate below $\lvert b\rvert = .02$ (Figure 4, Panel B). This is not a failure of the design to see time. In the same models the autoregressive effects were large and highly significant (pain $b = .395$, interference $b = .352$,

Figure 3. Constituent decomposition of the flow composite.



Note. Panel A shows each constituent's association with momentary pain when it is the only appraisal in the model (hollow) and when all four are entered together (filled), with the composite marked. Panel B shows the joint coefficients for all four pain measures. Panel C shows how far each coefficient moves between the two models. Panel D shows the AIC gained over the composite by splitting it in two and by using all four constituents. Panel E shows the within-person correlations among the constituents that produce the suppression. Panel F plots every parameter of the benchmark temporal and contemporaneous networks against the same parameter after the composite is substituted for the absorption node. Error bars are 95% confidence intervals.

threat b = .321, attention b = .266, all p < .001), so the models detect temporal structure where it exists and detect none between flow and pain (Figure S4, Panel D). The same held inside the flow construct itself: the challenge-skill condition at $t - 1$ did not predict the flow experience at t (elevation b = .052, p = .062; balance b = −.007, p = .70), which is 6% of the same-prompt elevation effect estimated on the identical rows (b = .856; Figure S4, Panels A and B, Table S6). The flow-pain relation in these data is co-occurrence, not prediction, and without temporal precedence the direction cannot be identified: absorption may dampen pain, or low pain may make absorption possible. H4 was not supported.

The channel comparison inverted the predicted ordering. Against apathy, the conventional reference, flow looked good on every measure (pain b = −.179, interference b = −.173, attention b = −.169, threat b = −.070), but relaxation looked better still (pain b = −.210, interference b = −.272, attention b = −.280, threat b = −.167; Figure 5, Panel D, Table 5). Tested head to head with flow as the reference, relaxation was reliably better on pain interference (b = −.098, p = .041), attention to pain (b = −.111, p = .005), and the threat value of pain (b = −.097, p = .001), and indistinguishable from flow on pain intensity (b = −.032, p = .42); anxiety and apathy were reliably worse than flow on all four (Figure 5, Panel E). Plotted over the full challenge-skill plane, the least painful region was the high-skill, low-challenge corner rather than the high-high corner (Figure 5, Panels B and C). This follows directly from the decomposition: within persons, skill tracked lower pain while challenge did not, and once entered jointly challenge tracked higher pain, so the most favourable corner of the plane is low-challenge and high-skill by construction (Figure 5, Panel F). H5 was therefore rejected in favour of the relaxation channel on

Table 3. Constituent decomposition of the flow composite.

Panel A. Each appraisal entered alone, predicting the four momentary pain measures.

Constituent	Pain intensity	Pain interference	Attention to pain	Threat value
Absorption	-0.069*** (0.014)	-0.089*** (0.017)	-0.048*** (0.013)	-0.039*** (0.009)
Enjoyment	-0.077*** (0.016)	-0.118*** (0.021)	-0.074*** (0.015)	-0.066*** (0.012)
Challenge	-0.004 (0.017)	+0.004 (0.022)	+0.006 (0.015)	+0.017 (0.012)
Skill	-0.088*** (0.016)	-0.125*** (0.021)	-0.086*** (0.017)	-0.059*** (0.013)
Flow experience (composite)	-0.096*** (0.018)	-0.133*** (0.022)	-0.077*** (0.016)	-0.068*** (0.012)

Panel B. The same four appraisals entered jointly.

Constituent	Pain intensity	Pain interference	Attention to pain	Threat value
Absorption	-0.055*** (0.011)	-0.062*** (0.014)	-0.029* (0.011)	-0.023** (0.009)
Enjoyment	-0.023 (0.013)	-0.057*** (0.016)	-0.035** (0.013)	-0.044*** (0.010)
Challenge	+0.032** (0.011)	+0.053*** (0.013)	+0.031** (0.011)	+0.035*** (0.008)
Skill	-0.045*** (0.013)	-0.063*** (0.016)	-0.049*** (0.013)	-0.026* (0.010)

Panel C. Nested likelihood-ratio tests of the composite against its constituents.

Outcome	Comparison	Δ df	χ^2	p	Δ AIC
Pain intensity	Composite vs. absorption + enjoyment	1	0.02	.891	-2.0
Pain intensity	Absorption + enjoyment vs. all four	2	20.98	<.001	17.0
Pain intensity	Composite vs. all four constituents	3	21.00	<.001	15.0
Pain interference	Composite vs. absorption + enjoyment	1	2.64	.104	0.6
Pain interference	Absorption + enjoyment vs. all four	2	33.49	<.001	29.5
Pain interference	Composite vs. all four constituents	3	36.13	<.001	30.1
Attention to pain	Composite vs. absorption + enjoyment	1	3.49	.062	1.5
Attention to pain	Absorption + enjoyment vs. all four	2	22.96	<.001	19.0
Attention to pain	Composite vs. all four constituents	3	26.45	<.001	20.5
Threat value	Composite vs. absorption + enjoyment	1	8.74	.003	6.7
Threat value	Absorption + enjoyment vs. all four	2	25.30	<.001	21.3
Threat value	Composite vs. all four constituents	3	34.04	<.001	28.0

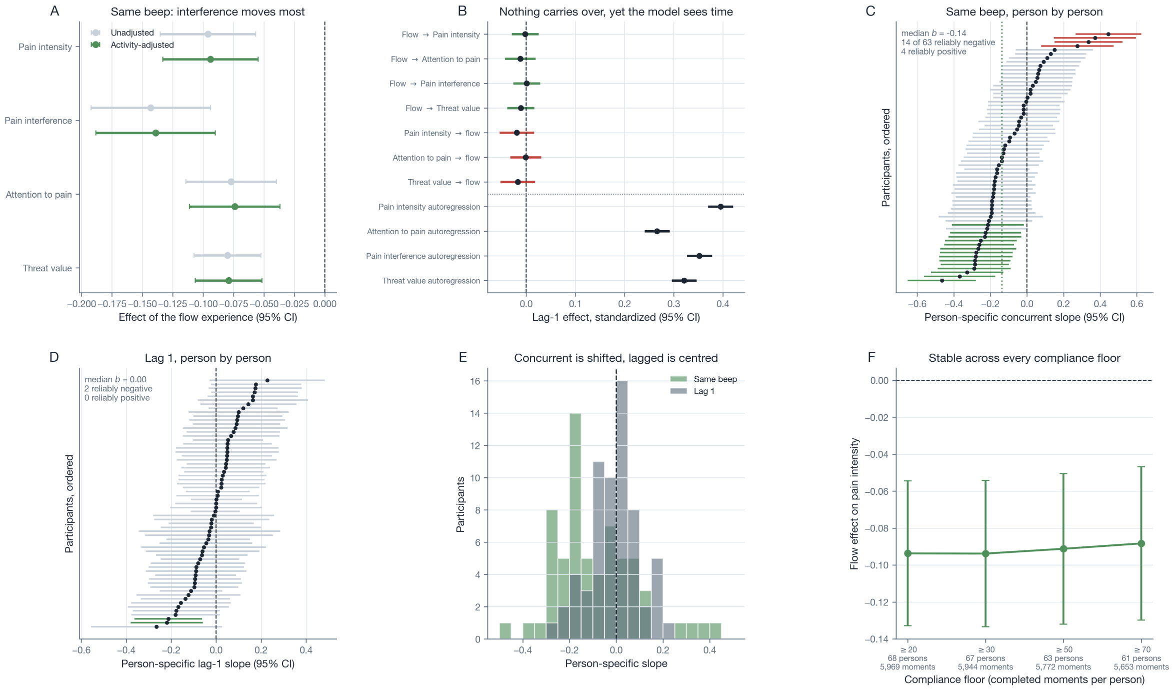
Note. Within-person (person-mean centred) predictors and outcomes. In Panel A "alone" means the constituent is the only appraisal in the model and "joint" means all four are entered together; the bottom row gives the composite for reference. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. Panel B compares nested fixed-effect structures under maximum likelihood with a random intercept and momentary physical activation as a covariate; the composite is the constituent model under the constraint that absorption and enjoyment carry equal weight and that challenge and skill carry none. The four appraisals are single momentary items; every flow variable is a secondary, analyst-constructed measure, since the source study analysed the items separately.

three of four outcomes and was not supported on the fourth.

3.5 Person-specific heterogeneity

The average concurrent association summarized wide variation, supporting H6. The person-specific flow-pain slope was negative in 73% of the 63 participants with enough moments to estimate it, with a median of $-.137$ and a range from $-.465$ to $+.445$; it was reliably negative in 14 participants and reliably positive in 4 (Figure 4, Panels C and E, Table 4). Those four are not noise around a common effect but participants whose confidence intervals exclude zero on the other side, and for whom a flow-based intervention would plausibly be counterproductive. The lag-1 slopes, by contrast, were centred on zero (median = .000, 49% negative, reliably negative in 2 and reliably positive in none), reproducing the pooled null at the individual level (Figure 4, Panel D). Between-person spread was of the same order as the average effect itself: the random-slope standard deviation was .114 for pain intensity against a fixed effect of $-.094$, and .146 for

Figure 4. The flow experience and the momentary experience of pain.



Note. Panel A shows the concurrent association of the flow experience with the four momentary pain measures, unadjusted and adjusted for physical activation. Panel B shows every lag-1 effect in both directions, with the four autoregressive effects from the same models below the dotted line as the reference for what a detectable temporal effect looks like here. Panels C and D show the person-specific concurrent and lag-1 slopes with their 95% confidence intervals, ordered by size; intervals excluding zero are highlighted. Panel E overlays the two person-level distributions. Panel F shows the concurrent estimate at four compliance floors.

interference against −.139 (Figure S6, Panel E).

Person-level flow variables related to the baseline profiles in a way that did not extend to the coupling (Figure S5, Table S9). A stronger baseline threat-value profile went with a lower average flow experience ($r = -.27$, $p = .028$; partial $r = -.27$, $p = .025$ controlling mean pain), as did higher neuroticism ($r = -.26$, $p = .035$; partial $r = -.29$, $p = .022$), whereas the biomedical profile was unrelated ($r = .03$, $p = .80$). No profile predicted the person-specific flow-pain slope (all $p \geq .18$). Baseline characteristics thus indexed how much flow a patient typically experienced but not how that experience related to their pain, so the heterogeneity in the coupling is not reducible to the trait differences measured here.

3.6 Robustness

The conclusions did not depend on the analytic choices that could have driven them (Figure S6, Table S10). The concurrent flow-pain estimate moved from −.094 to −.088 across compliance floors of 20, 30, 50, and 70 completed moments per person and remained significant throughout. Adding day and prompt position as level-1 covariates left every focal estimate essentially unchanged (elevation .744 to .746; flow to pain −.094 to −.101), although prompt position was itself a significant predictor. The condition results held on all four standardization metrics. Listwise deletion removed 4.6% of prompts; the removed moments carried slightly more negative affect ($d = 0.19$) and threat ($d = 0.15$) and slightly less physical activation ($d = -0.24$), but did not differ in pain intensity ($d = -0.12$, $p = .05$), interference, or attention (Figure S1, Panel D).

Table 4. The flow experience and the momentary experience of pain.

Panel A. The flow experience and the momentary experience of pain at the same beep.

Outcome	*b* unadjusted	SE	*b* adjusted	SE	95% CI	*p*	Random SD
Pain intensity	-0.096	0.020	-0.094	0.020	[-0.133, -0.055]	<.001	0.114
Pain interference	-0.143	0.025	-0.139	0.025	[-0.188, -0.090]	<.001	0.146
Attention to pain	-0.077	0.019	-0.074	0.019	[-0.111, -0.037]	<.001	0.097
Threat value	-0.080	0.014	-0.079	0.014	[-0.106, -0.052]	<.001	0.071

Panel B. Lag-1 effects in both directions, with the outcome's own autoregression in the same model.

Path	*b*	SE	*p*	Autoregression of the outcome
Flow → Pain intensity	-0.002	0.014	.888	0.395
Flow → Pain interference	0.001	0.014	.921	0.352
Flow → Attention to pain	-0.012	0.016	.451	0.266
Flow → Threat value	-0.011	0.014	.439	0.321
Pain intensity → flow	-0.019	0.018	.280	0.128
Attention to pain → flow	-0.001	0.016	.937	0.131
Threat value → flow	-0.017	0.018	.342	0.129

Panel C. Distribution of the person-specific flow-pain slope; "rel." marks slopes whose 95% interval excludes zero.

Slope	*n*	Median	IQR	Range	% neg.	Rel. neg.	Rel. pos.
Same beep	63	-0.137	[-0.20, 0.01]	[-0.47, 0.45]	73.0	14	4
Lag 1	63	0.000	[-0.08, 0.05]	[-0.27, 0.23]	49.2	2	0

Note. Panel A models are multilevel with a random slope for the flow experience and adjust for the challenge-skill condition; the adjusted column adds momentary physical activation. Panel B is estimated on person-standardized scores with lags built within day, so no lag crosses the overnight gap. Panel C reports unregularized per-person ordinary least squares estimates for participants with at least 50 usable moments; "reliably" means the 95% confidence interval excludes zero. The four appraisals are single momentary items; every flow variable is a secondary, analyst-constructed measure, since the source study analysed the items separately.

Table 5. The four challenge-skill channels and the momentary experience of pain.

Panel A. Each channel against apathy, the conventional reference.

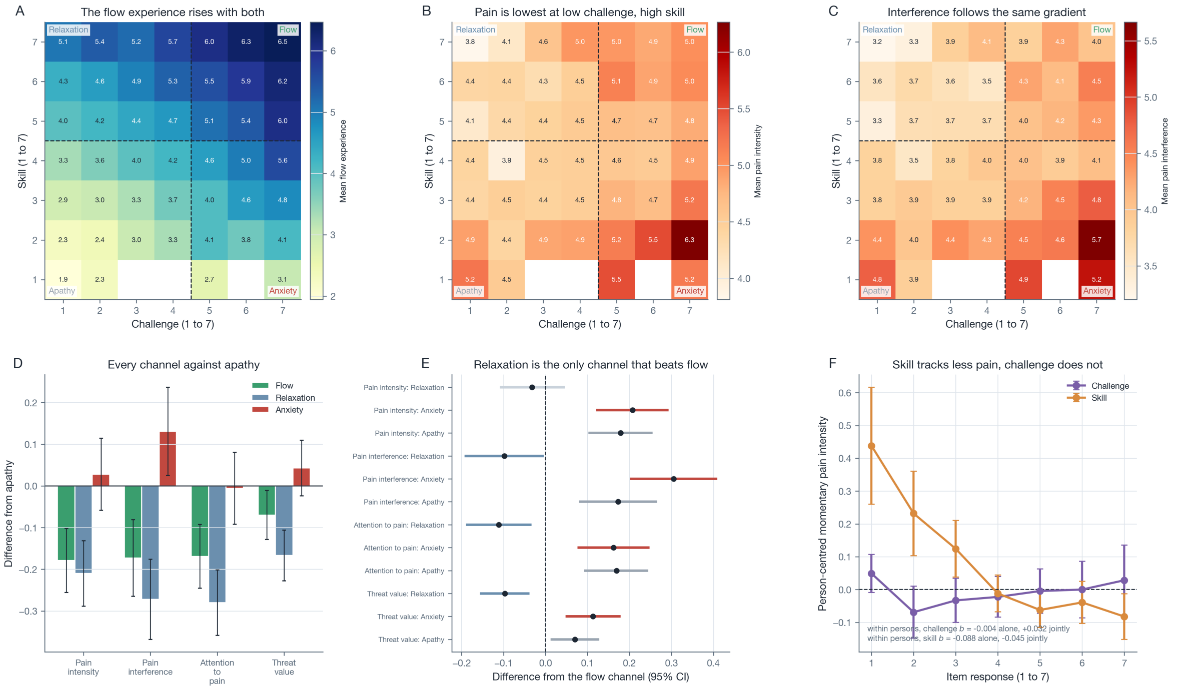
Channel	Pain intensity	Pain interference	Attention to pain	Threat value
Flow	-0.179 (<.001)	-0.173 (<.001)	-0.169 (<.001)	-0.070 (.018)
Relaxation	-0.210 (<.001)	-0.272 (<.001)	-0.280 (<.001)	-0.167 (<.001)
Anxiety	+0.028 (.522)	+0.131 (.015)	-0.006 (.888)	+0.043 (.201)

Panel B. Each channel against flow, the direct head-to-head comparison.

Channel	Pain intensity	Pain interference	Attention to pain	Threat value
Relaxation	-0.032 (.425)	-0.098 (.041)	-0.111 (.005)	-0.097 (.001)
Anxiety	+0.207 (<.001)	+0.305 (<.001)	+0.162 (<.001)	+0.113 (<.001)
Apathy	+0.179 (<.001)	+0.173 (<.001)	+0.169 (<.001)	+0.070 (.018)

Note. Channels are formed from person-mean centred challenge and skill: flow is both above the person's own mean, relaxation skill only, anxiety challenge only, and apathy neither. Entries are differences on the raw 1 to 7 response scale from multilevel models with a random person intercept, with p in parentheses. Negative values mean less pain, less interference, less attention to pain, or a lower threat value than the reference channel. The four appraisals are single momentary items; every flow variable is a secondary, analyst-constructed measure, since the source study analysed the items separately.

Figure 5. The challenge-skill plane and where the least painful moments sit.



Note. Panels A to C show the mean flow experience, mean pain intensity, and mean pain interference across the full raw challenge-skill plane, with the four channels marked and cells holding fewer than 15 moments left blank. Panel D shows each channel against apathy, the conventional reference. Panel E shows each channel against flow, the direct head-to-head test; grey intervals are not significant. Panel F shows the person-centred mean pain at each level of challenge and of skill. Error bars and intervals are 95% confidence intervals.

4 Discussion

This study asked whether the flow construct survives contact with the daily life of patients with chronic pain, and whether the momentary flow experience accompanies or precedes a better experience of pain. Three findings stand out. First, the challenge-skill condition acted additively rather than configurationally: no functional form of the match between challenge and skill improved on a model with the two components entered separately, and between persons better-matched activity went with less flow experience, not more. Second, the composite lost information relative to its constituents, because absorption and skill tracked lower pain while challenge, once the others were held constant, tracked higher pain. Third, the flow experience accompanied a better experience of pain at the same moment, most strongly for interference, but predicted nothing at the next prompt, and the least painful region of the challenge-skill plane was the relaxation channel rather than flow. Each research question is discussed in turn, followed by limitations and future directions.

4.1 Main findings

4.1.1 Additive rather than configural (RQ1)

The classic account is explicitly configural: flow is not what high challenge produces, nor what high skill produces, but what their correspondence at a high level produces (Csikszentmihalyi, 1990; Massimini and Carli, 1988). In these data that claim failed in every form in which it was tested. The balance term was null within persons and reversed between them; a one-sided kink at equality, a multiplicative interaction, and a full second-order response surface each added nothing; and the four-channel quadrant scheme improved fit only marginally and was rejected by BIC. What remained was an ordinary additive structure in which skill weighed about twice as much as challenge.

Two features of this result deserve emphasis. The first is that the null is informative rather than underpowered. With just under 6,000 moments the balance estimate was two orders of magnitude smaller than the elevation effect it competed with and was pinned down to within ±.04, so the finding is that there is nothing to see, not that nothing could be seen (Lakens et al., 2018). The second is that the tests do not depend on the assumption that makes difference scores fragile. A balance term is interpretable only if a point on the challenge scale is commensurate with a point on the skill scale, which two single diary items cannot guarantee; the interaction and response-surface tests keep both components in their own metric and require no such assumption (Edwards and Parry, 1993; Edwards, 1994). They agreed with the balance test, and the descriptive picture, flat mean flow experience across the whole range of mismatch within a band of elevation, agreed with both.

The between-person reversal is the one positive finding in this section, and it has a clinical reading. A patient whose activities are consistently well matched in challenge and skill is plausibly a patient who has adjusted downward to what feels safely manageable. Challenge and skill then correspond nicely, but at a low level, and activity avoidance scores statistically as good balance. That interpretation connects directly to the literature on pacing, avoidance, and endurance in chronic pain (Andrews et al., 2012; Vlaeyen and Linton, 2000), and it is a pattern one would not expect to see in the student and worker samples in which the channel model was developed (Carli et al., 1988). It also suggests that the elevation term, which indexes the level of engagement, is the one that actually varies meaningfully in this population.

These results sit alongside the wider flow literature rather than against it. In meta-analysis the balance between challenge and skill relates to flow only modestly (Fong et al., 2015), and experimental work has found the balance effect to be conditional rather than general (Engeser and Rheinberg, 2008). The present contribution is to test the claim on its own home ground, signal-contingent experience sampling, in a clinical sample and in several functional forms at once.

4.1.2 The composite adds a label where the components carry the information (RQ2)

VanderWeele (2022) distinguishes three worlds behind an identical composite coefficient, and these data fall into the least comfortable of them. The composite of all four appraisals averages over constituents that pull in opposite directions with respect to pain: absorption and skill favourable, enjoyment redundant once the other two are present, and challenge unfavourable. Reporting the composite alone would have hidden both the driving constituent and the counteracting one, and would have obscured the fact that the most clinically actionable component, challenge, is the one that carries a cost.

That challenge carries a cost is substantively plausible rather than a statistical curiosity. A demanding moment takes something out of a person with chronic pain even when they are equal to it, and in this sample challenge co-occurred with physical exertion. The suppression pattern is exactly what should be expected when a variable with an unfavourable direct relation is positively correlated with variables that have favourable ones (MacKinnon et al., 2000), and it converges with the channel finding: if challenge is unfavourable and skill favourable, the best corner of the plane must be low-challenge and high-skill, which is what the surfaces show.

The network substitution makes the same point on a different axis. The decomposition says the composite is carried by two of its four parts; the network says that in a dynamic system the composite behaves indistinguishably from its dominant constituent, reproducing 22 parameters at $r = .997$ with no change of sign or significance. Two independent lines of evidence, one coefficient-level and one multivariate, converge on the conclusion that the flow architecture adds a label rather than information. The recommendation that follows is concrete and transferable: when flow is measured in experience sampling, report the constituents separately and jointly alongside any composite, because the composite can conceal a sign reversal among them.

4.1.3 Co-occurrence rather than analgesia (RQ3)

The concurrent association was robust, survived adjustment for physical activation, and was ordered as an attentional account predicts, with interference moving more than intensity (Eccleston and Crombez, 1999; Van Damme et al., 2010). But it did not extend in time. Neither direction produced a lag-1 effect, while the autoregressions in the same models were large, so the null is informative about this design rather than uninterpretable. Absent temporal precedence, the relation is co-occurrence and the direction is not identified: absorption may dampen pain, and low pain may equally make absorption possible.

That is weaker evidence than the flow-induced-analgesia proposal requires, and it is an honest result rather than a disappointing one. The induced-flow paradigms that motivate the proposal measure pain during an induction (Gromakovskis and Gūtmanis, 2025); the present design asks whether naturally occurring absorption still shows a footprint an hour and a half later, and it does not. The two findings are compatible, and the gap between them is exactly the ecological question worth posing.

The channel result is the most directly contrary to the flow account, and also the most clinically useful. If the goal is a less painful moment in daily life, these data point toward being competent at something undemanding rather than toward seeking out optimal challenge. Relaxation beat flow on interference, attention, and threat, and matched it on intensity; anxiety and apathy were worse than both. This is a moderate, empirically grounded counterweight to the enthusiasm for flow-based intervention: not that flow does nothing, but that in everyday life it is the skill component that does the work, the challenge component carries a cost, and the most favourable moments are not the flow moments.

Finally, the group average was not universal. In 4 of 63 participants the concurrent flow-pain slope was reliably positive, and no baseline profile predicted where a given patient fell. Baseline characteristics indexed how much flow a patient typically experienced but not how that experience related to their pain, a dissociation between level and coupling that is an argument for measuring the dynamics directly rather than inferring them from questionnaires (Wright and Woods, 2020; Fisher et al., 2018; Molenaar, 2004).

4.2 Limitations

Several limitations qualify these conclusions. They concern, in turn, the measurement of the construct, the causal status of the associations, the temporal resolution, the confounding of challenge with exertion, the statistical power for person-level analyses, and the generality of the sample.

First, and most importantly, the flow measure rests on four single momentary items and covers three of the nine dimensions Csikszentmihalyi describes, leaving clear goals, unambiguous feedback, sense of control, loss of self-consciousness, and time transformation unmeasured. It is a core flow proxy rather than a flow scale (Jackson and Eklund, 2002), and it is a secondary, analyst-constructed measure, since the source study analysed these items separately (Viane et al., 2004). Single-item measurement also raises a specific technical objection to the decomposition: measurement error in correlated predictors can produce sign reversals under joint entry as an artefact. Three considerations weigh against that reading here. The reversal is not marginal ($p = .004$ in nearly 6,000 moments) and holds on all four pain outcomes; it is theoretically interpretable, since demanding activity plausibly costs something; and it converges with the independent channel finding, which does not depend on joint entry at all. The objection nevertheless remains open, and replication with multi-item momentary flow scales is the way to close it.

Second, the analysis is observational. Nothing was manipulated, and flow, pain, and activity can influence one another through unmeasured time-varying causes such as mood, medication, or the demands of the moment, so all estimates are within-person predictive associations rather than causal effects (Rohrer, 2018). The channel comparison in particular contrasts moments that participants selected rather than conditions to which they were assigned.

Third, the temporal resolution is coarse and unequal. Adjacent prompts are separated by variable hours, no lag is carried overnight, and momentary reports are retrospective over a short window. Flow is by definition a state that fluctuates from moment to moment, and an episode can begin and end entirely within one interval. The absence of lagged effects is therefore open to two readings that these data cannot separate: a measurement limitation, in which the state fluctuates faster than the sampling grid, or a substantive null, in which flow accompanies less pain only while it lasts. The first is the more likely and is how the result is framed here, but the second is compatible with the induction studies, which also measure pain during rather than after the state (Hamaker and Wichers, 2017).

Fourth, the confounding of challenge with physical exertion could be adjusted for but not dissolved. Challenge correlated $r = .57$ within persons with self-reported physical activation, and although every model was estimated with activation as a covariate and none of the focal estimates moved, the open-text activity descriptions were not coded, so demand and exertion could not be separated substantively. This limits how far the channel ordering can be interpreted as a statement about psychological demand rather than about physical load.

Fifth, the person-level analyses rest on 68 participants, and 63 of them contributed enough moments for an individual slope. That is adequate for describing heterogeneity but modest for explaining it, so the absence of trait moderation should be read as a failure to detect rather than as evidence of no moderation.

Sixth, all data come from a single chronic-pain cohort recruited at one pain clinic, with a predominance of women and of back and widespread pain. The channel model was developed in non-clinical samples, and the between-person balance reversal reported here may well be specific to a population that has reason to adjust its activities downward. Replication in other pain conditions, and in non-clinical samples using the same nested tests, is needed before either the structural or the substantive conclusions are generalized.

4.3 Future directions

Four directions follow. First, the structural tests used here cost little and should be routine. Any study that reports a momentary flow composite can also report its constituents alone and jointly, and any study that reports a balance term can also report the nested comparison against an additive model and a response surface. Whether the failure of configurality found here is specific to chronic pain or general is an empirical question that existing datasets could answer immediately.

Second, the temporal question needs a design built for it. Burst sampling, event-contingent prompts triggered by the start and end of activities, or continuous passive sensing of activity would shorten the interval enough to distinguish a genuinely concurrent-only relation from one that the ninety-minute grid cannot resolve. Only such a design can move the flow-induced-analgesia question from co-occurrence toward precedence.

Third, the channel result invites an experimental test. Because relaxation, not flow, was the least painful region of the plane, a within-person experiment that assigns competent but undemanding activity against equally absorbing but demanding activity would separate the skill and challenge components that these observational data can only decompose statistically. That is also the design that would test whether the cost of challenge is psychological demand or physical exertion.

Fourth, the dissociation between level and coupling deserves following up. Baseline profiles predicted how much flow a patient experienced but not how that experience related to their pain, and four patients showed a reliably positive coupling. Whether person-specific coupling estimates carry information beyond questionnaires, and whether they can be used to decide for whom an activity-based intervention is appropriate, is an open and clinically consequential question (Wright and Woods, 2020).

5 Conclusion

Measured moment by moment in the daily life of patients with chronic pain, the flow construct did not hold together in the way its standard operationalization assumes: the match between challenge and skill added nothing to the two components entered separately, in any functional form tested, and between persons better-matched activity accompanied less flow rather than more, while the four-item composite lost information relative to its constituents because absorption and skill tracked lower pain and challenge, once the others were held constant, tracked higher pain. The flow experience did accompany a better experience of pain at the same prompt, most strongly its interference, but it predicted nothing at the next prompt while autoregressive effects in the same models remained large, and the least painful region of the challenge-skill plane was the low-challenge, high-skill relaxation channel rather than flow. Taken together, these results recommend that experience-sampling research report the constituents of flow separately and jointly rather than a composite, and they suggest that what accompanies relief in everyday life with chronic pain is competence without demand rather than optimal challenge.

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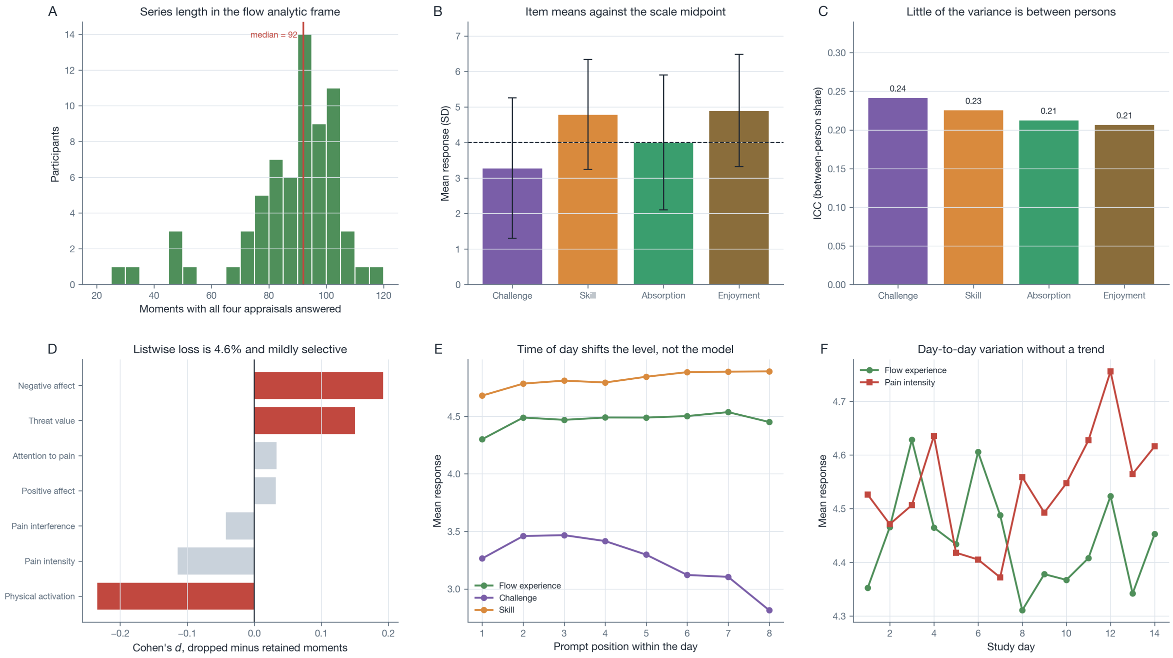
Appendix

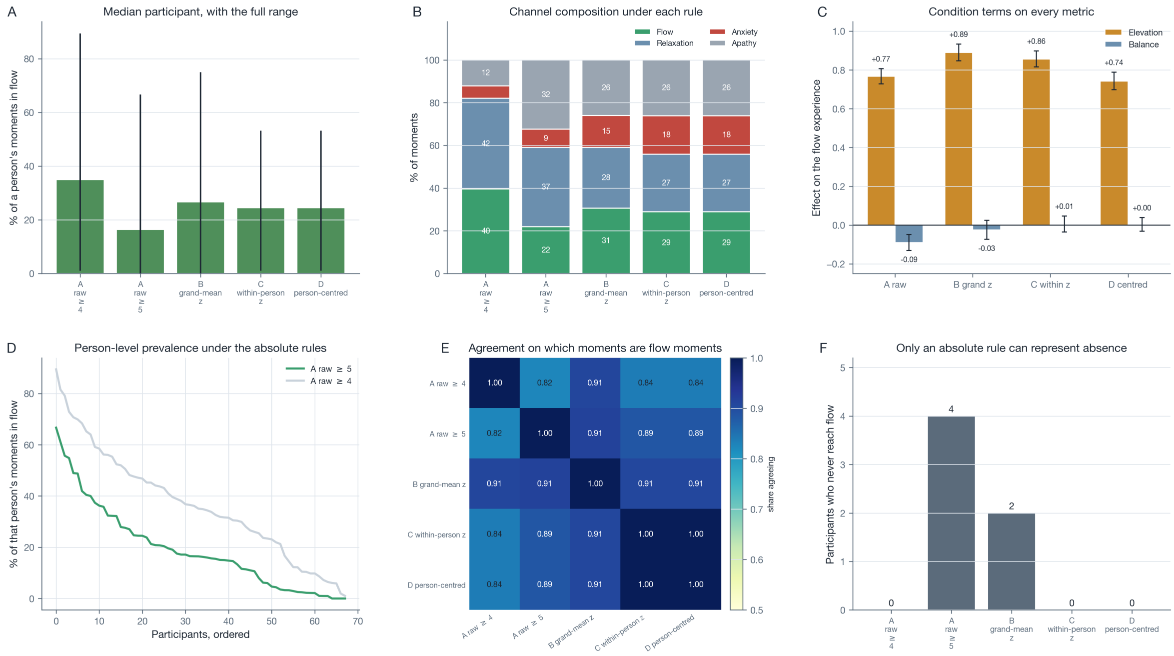
Supplementary Figures

Figure S1. The flow analytic frame.



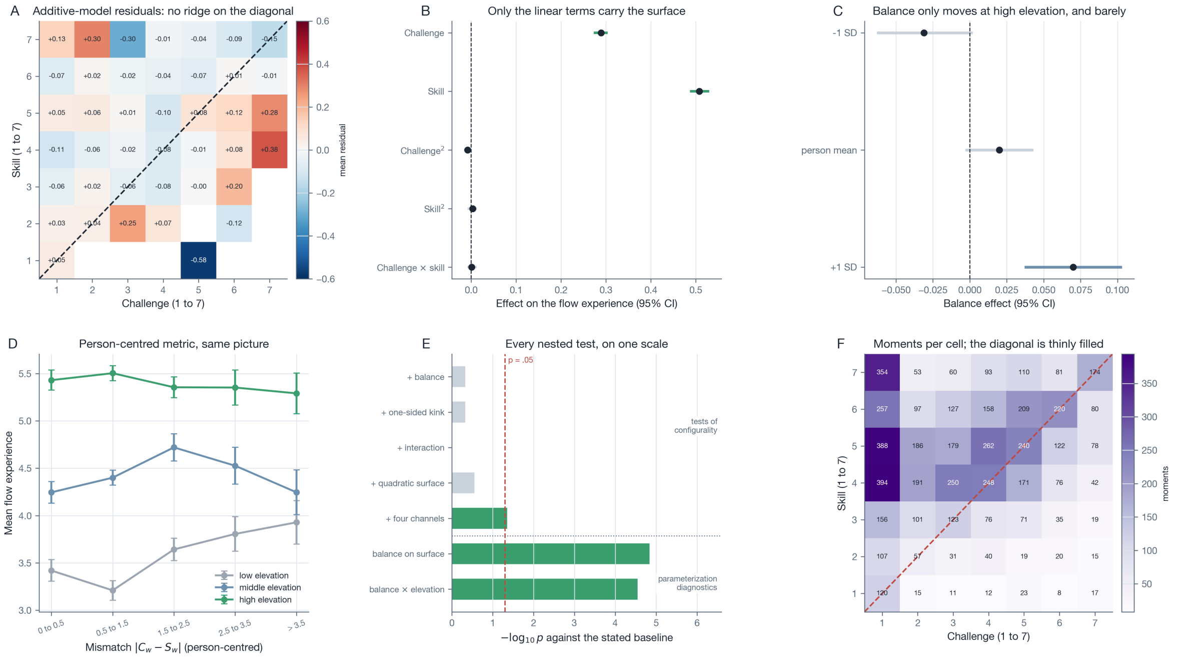
Note. Panel A shows the number of moments per participant with all four appraisals answered. Panel B shows the item means with their standard deviations against the scale midpoint. Panel C shows the between-person variance share of each item. Panel D compares the moments dropped by listwise deletion with those retained; coloured bars are significant at $p < .05$. Panel E shows the item means by prompt position within the day, restricted to positions with at least 150 moments. Panel F shows the flow experience and pain intensity across the 14 study days.

Figure S2. How the standardization rule decides every descriptive claim.



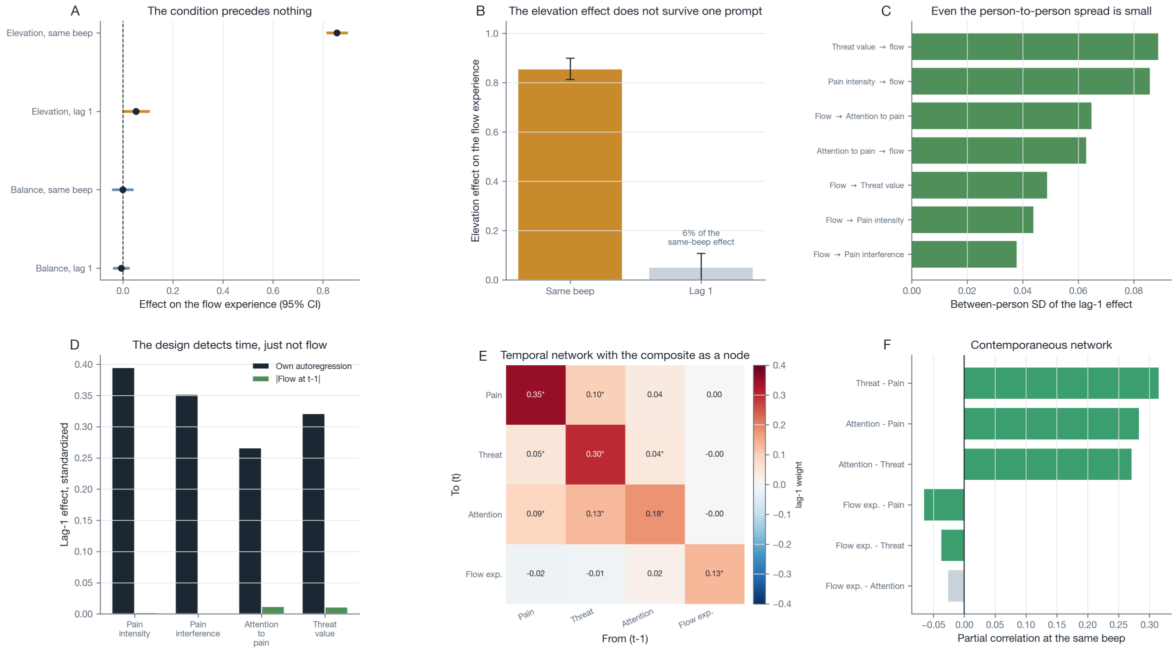
Note. Panel A shows the median participant's share of flow moments under each rule, with the full person range. Panel B shows the channel composition of all moments under each rule. Panel C shows the balance and elevation effects estimated on each metric. Panel D shows the ordered per-person prevalence under the two absolute rules. Panel E shows how often two rules agree on whether a given moment is a flow moment. Panel F shows how many participants never reach flow under each rule; only the absolute rules can represent absence.

Figure S3. Diagnostics of the balance parameterization.



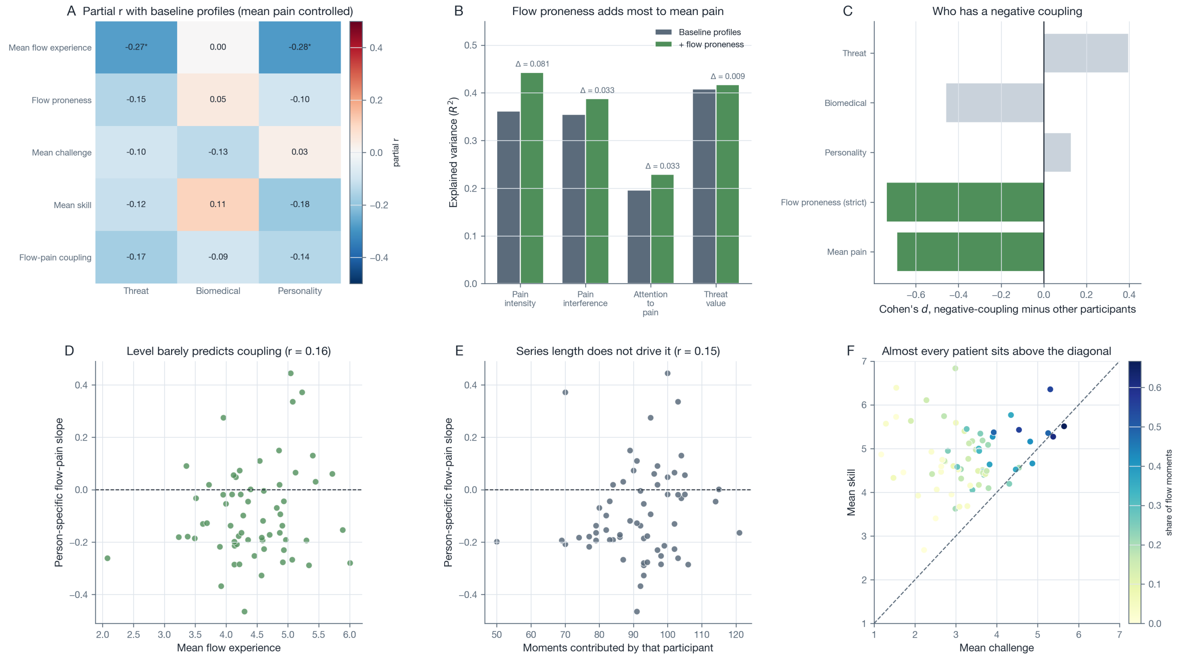
Note. Panel A shows the mean residual of the additive challenge-plus-skill model over the challenge-skill plane; a configural effect would appear as a ridge along the dashed diagonal. Panel B shows the coefficients of the full second-order response surface. Panel C shows the simple slopes of balance at low, average, and high elevation inside the balance-and-elevation parameterization. Panel D repeats the mismatch-within-elevation description on the person-centred metric. Panel E places every nested test on a common significance scale, separating the tests of configurality from the two diagnostics of the parameterization. Panel F shows the number of moments in each cell of the plane.

Figure S4. Temporal architecture of the construct.



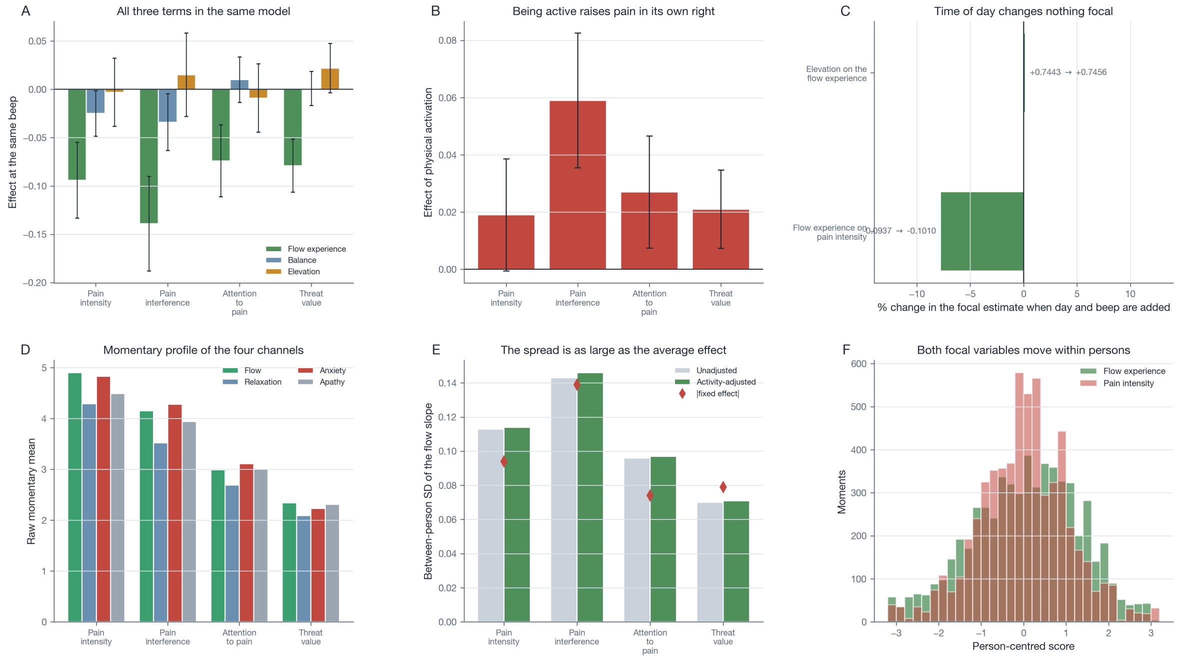
Note. Panel A shows the condition terms predicting the flow experience at the same prompt and one prompt later, estimated on the same rows. Panel B compares the same-prompt and lagged elevation effects directly. Panel C shows the between-person standard deviation of each lag-1 effect. Panel D places the lag-1 flow effects next to the outcome's own autoregression from the same model. Panels E and F show the temporal and contemporaneous networks estimated with the flow composite in place of the absorption node; asterisks mark $p < .05$.

Figure S5. Person-level flow proneness, coupling, and baseline profiles.



Note. Panel A shows the partial correlation of each person-level flow variable with each baseline profile, controlling the participant's mean momentary pain; asterisks mark $p < .05$. Panel B shows the variance in each mean momentary state added by flow proneness over the three profiles. Panel C compares participants whose flow-pain slope is negative with the rest. Panels D and E plot the person-specific slope against the participant's mean flow experience and against the number of moments they contributed. Panel F places each participant on the challenge-skill plane, coloured by their share of flow moments.

Figure S6. Robustness of the flow-pain association.



Note. Panel A shows the flow experience, balance, and elevation estimated together in the same concurrent model. Panel B shows the effect of momentary physical activation in those models. Panel C shows how far each focal estimate moves when day and prompt position are added as covariates. Panel D shows the raw momentary profile of the four channels. Panel E compares the between-person standard deviation of the flow slope with the absolute size of the fixed effect. Panel F shows the person-centred distributions of the two focal variables.

Supplementary Tables

Table S1. Correlations among the four appraisals and the composite.

Item	Challenge	Skill	Absorption	Enjoyment	Flow experience
Challenge		0.14	0.64	0.21	0.53
Skill	0.17		0.41	0.68	0.62
Absorption	0.53	0.42		0.42	0.88
Enjoyment	0.23	0.59	0.52		0.81
Flow experience	0.45	0.57	0.90	0.85	

Note. Below the diagonal: within-person correlations after person-mean centring. Above the diagonal: between-person correlations of the person means. The flow experience is the mean of absorption and enjoyment, so its correlations with those two items are inflated by construction.

Table S2. Prevalence of flow moments under each classification rule.

Rule	Condition met	Gated flow	Median person	Person range	At zero
A. Raw, both items ≥ 4	39.6	36.8	35.0	1.0 to 89.4	0/68
A. Raw, both items ≥ 5	22.0	19.3	16.4	0.0 to 66.7	4/68
B. Grand-mean z	30.6	27.8	26.7	0.0 to 75.0	2/68
C. Within-person z	28.9	24.6	24.5	1.0 to 53.2	0/68
D. Person-mean centred	28.9	24.6	24.5	1.0 to 53.2	0/68

Note. All shares are percentages of moments, except the last column, which counts participants who never reach flow. The gated criterion requires both the challenge-skill condition and a flow experience above the same cut. Relative rules force a within-person distribution and therefore assign above-average moments to every participant, which is why only the absolute rules can represent the absence of flow and why every prevalence claim in the text is made on the absolute metric.

Table S3. Condition terms predicting the flow experience on every standardization metric.

Metric	Term	b	SE	95% CI	p
D. Person-mean centered (primary)	Balance	0.004	0.018	[-0.031, 0.039]	.826
D. Person-mean centered (primary)	Elevation	0.743	0.023	[0.698, 0.788]	<.001
C. Within-person z	Balance	0.005	0.021	[-0.036, 0.046]	.806
C. Within-person z	Elevation	0.857	0.021	[0.816, 0.898]	<.001
B. Grand-mean z	Balance	-0.025	0.025	[-0.074, 0.024]	.315
B. Grand-mean z	Elevation	0.890	0.022	[0.847, 0.933]	<.001
A. Raw absolute scale	Balance	-0.090	0.021	[-0.131, -0.049]	<.001
A. Raw absolute scale	Elevation	0.768	0.020	[0.729, 0.807]	<.001

Note. Person-mean centring is the primary metric. Across all four metrics the elevation effect stays between .74 and .89 and the balance effect is never positive.

Table S4. Fixed effects of every nested condition model.

Model	Term	*b*	SE	95% CI	*p*
challenge + skill	Challenge	0.285	0.008	[0.270, 0.300]	<.001
challenge + skill	Skill	0.507	0.010	[0.488, 0.526]	<.001
challenge + skill + balance	Challenge	0.284	0.008	[0.269, 0.300]	<.001
challenge + skill + balance	Skill	0.508	0.010	[0.488, 0.527]	<.001
challenge + skill + balance	Balance	-0.009	0.011	[-0.031, 0.014]	.452
challenge + skill + overload	Challenge	0.276	0.014	[0.248, 0.304]	<.001
challenge + skill + overload	Skill	0.516	0.016	[0.485, 0.548]	<.001
challenge + skill + overload	Overload kink	0.017	0.023	[-0.028, 0.062]	.452
challenge x skill	Challenge	0.285	0.008	[0.270, 0.300]	<.001
challenge x skill	Skill	0.507	0.010	[0.488, 0.526]	<.001
challenge x skill	Challenge × skill	0.000	0.005	[-0.010, 0.010]	.970
quadratic response surface	Challenge	0.289	0.008	[0.273, 0.304]	<.001
quadratic response surface	Skill	0.508	0.011	[0.487, 0.530]	<.001
quadratic response surface	Challenge^2	-0.008	0.004	[-0.016, 0.000]	.051
quadratic response surface	Skill^2	0.003	0.005	[-0.008, 0.013]	.622
quadratic response surface	Challenge × skill	0.001	0.005	[-0.009, 0.012]	.804
quadratic surface + balance	Challenge	0.295	0.008	[0.279, 0.311]	<.001
quadratic surface + balance	Skill	0.503	0.011	[0.482, 0.525]	<.001
quadratic surface + balance	Challenge^2	-0.034	0.007	[-0.048, -0.020]	<.001
quadratic surface + balance	Skill^2	-0.019	0.007	[-0.033, -0.005]	.010
quadratic surface + balance	Balance	-0.130	0.030	[-0.188, -0.071]	<.001
quadratic surface + balance	Challenge × skill	0.048	0.012	[0.024, 0.071]	<.001
balance x elevation	Balance	0.020	0.012	[-0.003, 0.043]	.084
balance x elevation	Elevation	0.810	0.019	[0.773, 0.847]	<.001
balance x elevation	Balance × elevation	0.043	0.010	[0.023, 0.063]	<.001

Note. All models predict the flow experience from person-mean centred predictors with a random person intercept, estimated by maximum likelihood. The "quadratic surface + balance" model is reported for completeness; its balance coefficient is not interpretable in isolation because balance correlates .93 with the squared discrepancy that the surface already contains.

Table S5. Diagnostics of the balance parameterization.

Panel A. Why the balance term is not interpretable on its own.

Quantity	Value
r(balance, $-(C-S)^2$)	0.927
r(balance, C^2)	-0.506
r(balance, S^2)	-0.324
r(balance, elevation)	0.030
b challenge (additive model)	0.285
b skill (additive model)	0.507

Panel B. Simple slopes of balance across elevation.

Elevation	Value (centred)	Balance b	SE	95% CI	p
-1 SD	-1.179	-0.031	0.016	[-0.063, 0.002]	.064
person mean	0.000	0.020	0.012	[-0.003, 0.043]	.084
+1 SD	1.179	0.070	0.017	[0.037, 0.103]	<.001

Panel C. The two comparisons in which balance appears to matter.

Comparison	χ^2	df	p	Δ AIC
balance added to the quadratic surface	18.87	1	<.001	16.9
balance x elevation inside the balance-elevation form	17.60	1	<.001	15.6

Note. C is challenge and S skill, both person-mean centred. Balance is a monotone transformation of the absolute discrepancy and therefore nearly collinear with the squared discrepancy that a quadratic surface already contains; the balance-and-elevation form additionally constrains challenge and skill to equal weight, which the free additive estimates reject. Both comparisons in Panel C are properties of those parameterizations rather than evidence of configurality, which is why no configural form beats the additive model in Table 2.

Table S6. Does the challenge-skill condition precede the flow experience?

Model	Term	b	SE	p	n obs
Lag 1	Balance at $t-1$	-0.007	0.017	.696	5,065
Lag 1	Elevation at $t-1$	0.052	0.028	.062	5,065
Lag 1	Flow experience at $t-1$	0.108	0.019	<.001	5,065
Same beep	Balance, same beep	-0.001	0.022	.963	5,065
Same beep	Elevation, same beep	0.856	0.022	<.001	5,065

Note. Both models are fitted on the same rows so the lagged and same-beep estimates are directly comparable. The lag-1 model controls the flow experience's own autoregression; predictors are person-standardized and lags are built within day.

Table S7. Substituting the flow composite for the single absorption node.

Edge	Benchmark (absorption)	p	With the composite	p	Difference
Pain(t-1) → activity node(t)	-0.048	.065	-0.021	.413	0.027
Activity node(t-1) → Pain(t)	0.003	.809	0.000	.991	-0.003
Pain - activity node (contemporaneous)	-0.064	.003	-0.066	.002	-0.002
Attention - activity node (contemporaneous)	-0.022	.188	-0.027	.109	-0.005

Note. Both networks are multilevel vector autoregressive models on four nodes: momentary pain, its threat value, attention to pain, and either goal absorption (benchmark) or the flow composite. Only the edges that involve the substituted node can change; none changes sign or significance.

Table S8. Momentary profile of the four challenge-skill channels.

Channel	*n*	%	Pain	Interference	Attention	Threat	Flow exp.	Activation
Flow	1,314	22.0	4.90	4.15	2.99	2.34	5.79	4.33
Relaxation	2,214	37.0	4.29	3.52	2.69	2.09	4.64	2.47
Anxiety	516	8.6	4.83	4.28	3.11	2.23	4.44	4.44
Apathy	1,932	32.3	4.49	3.94	3.00	2.31	3.33	2.46

Note. Unadjusted momentary means on the raw 1 to 7 scale under the strict absolute rule (both condition items at least 5). Activation is the self-reported physical activation item, which is far higher in the two high-challenge channels and is the reason every model reported in the text is also estimated with activation as a covariate.

Table S9. Person-level flow proneness and the baseline profiles.

Panel A. Person-level flow variables against the baseline profiles.

Person-level flow variable	Baseline profile	*n*	*r*	*p*	Partial *r*	*p*
Flow proneness (strict)	Threat	67	-0.092	.459	-0.146	.238
Flow proneness (strict)	Biomedical	67	0.294	.016	0.048	.702
Flow proneness (strict)	Personality	64	0.045	.722	-0.097	.444
Mean flow experience	Threat	67	-0.268	.028	-0.274	.025
Mean flow experience	Biomedical	67	0.032	.798	0.003	.982
Mean flow experience	Personality	64	-0.264	.035	-0.285	.022
Mean challenge	Threat	67	-0.047	.707	-0.096	.440
Mean challenge	Biomedical	67	0.166	.178	-0.131	.290
Mean challenge	Personality	64	0.157	.216	0.030	.812
Mean skill	Threat	67	-0.114	.360	-0.117	.344
Mean skill	Biomedical	67	0.117	.345	0.113	.361
Mean skill	Personality	64	-0.172	.174	-0.185	.142
Flow-pain coupling	Threat	62	-0.119	.355	-0.171	.185
Flow-pain coupling	Biomedical	62	0.138	.284	-0.094	.466
Flow-pain coupling	Personality	59	-0.016	.905	-0.141	.287

Panel B. Variance in the mean momentary states added by flow proneness.

Mean momentary state	*n*	R^2 profiles	R^2 + flow proneness	ΔR^2
Attention to pain	64	0.196	0.229	0.033
Pain intensity	64	0.362	0.443	0.081
Pain interference	64	0.355	0.388	0.033
Threat value	64	0.408	0.417	0.009

Panel C. Participants whose flow-pain slope is negative against the rest.

Person-level variable	*n* negative	*n* other	Mean, negative	Mean, other	Cohen's *d*	*p*
Threat	45	17	0.059	-0.271	0.398	.151
Biomedical	45	17	-0.071	0.272	-0.460	.131
Personality	42	17	-0.002	-0.133	0.129	.654
Flow proneness (strict)	45	17	0.161	0.293	-0.739	.027
Mean pain	45	17	4.313	5.073	-0.690	.029

Note. Partial correlations control the participant's mean momentary pain. The threat profile combines the Pain Catastrophizing Scale and the Pain Vigilance and Awareness Questionnaire, the biomedical profile pain severity and duration, and the personality profile NEO-FFI neuroticism. Flow proneness is the share of a participant's moments meeting the strict absolute criterion. No profile predicts the person-specific flow-pain slope, which is the contrast the text draws with the level effects.

Table S10. Robustness of the flow analyses.

Panel A. The flow-pain estimate at four compliance floors.

Compliance floor	*n* persons	*n* moments	*b*	SE	*p*
≥ 20 moments	68	5,969	-0.0937	0.0200	<.001
≥ 30 moments	67	5,944	-0.0938	0.0202	<.001
≥ 50 moments	63	5,772	-0.0912	0.0208	<.001
≥ 70 moments	61	5,653	-0.0883	0.0212	<.001

Panel B. Day and prompt position added as level-1 covariates.

Model	Focal term	*b* without	*b* with	*p*	*p* beep	*p* day
V2: condition → experience	elevation_w	0.7443	0.7456	<.001	<.001	.959
Flow → pain (activity-adjusted)	FLOWEXP_w	-0.0937	-0.1010	<.001	<.001	.079

Panel C. Moments lost to listwise deletion.

Measure	Mean, retained	Mean, dropped	*n* dropped	Cohen's *d*	*p*
Pain intensity	4.53	4.36	286	-0.115	.052
Pain interference	3.86	3.79	272	-0.043	.476
Attention to pain	2.89	2.95	255	0.034	.593
Threat value	2.23	2.44	268	0.151	.019
Negative affect	1.92	2.14	280	0.193	.002
Positive affect	4.23	4.27	286	0.033	.583
Physical activation	3.05	2.62	274	-0.235	<.001

Note. Panel A repeats the activity-adjusted concurrent model of pain intensity with a stricter minimum number of completed moments per participant. In Panel B the focal estimate is compared with and without day and prompt position; prompt position is itself significant but moves no focal estimate. Panel C compares the moments retained in the flow analytic frame with the 4.6% dropped for a missing appraisal.